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language translation by Allan Bloom–shows the intensity of Kojève's study and thought and the depth of his insight into Hegel's Phenomenology. More important–for Kojève was above all a philosopher and not an ideologue–this profound and venturesome work on Hegel will expose the readers to the excitement of discovering a great mind in all its force and power.} }

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Machinagogy: Experiments in Staging Teaching Dramas with LLMs

The dizzying world of generative AI is marked by a relentless simulation of existing social relations. AI serves customers; writes code and produces digital assets for the knowledge economy; counsels those who desire or need alternatives to human illness and therapy. And it teaches. In the three or so years since OpenAI's ChatGPT was launched in November 2022, it has become almost superfluous to seek to improve upon the pedagogical capabilities of the so-called foundational models themselves: ChatGPT, Anthropic's Claude, Google's Gemini. As of 2026, these models can perform textual tasks - simplify, summarize, translate, analyze, quiz - that translate into digitized pedagogy.

However AI tutors remained locked and limited in important ways. This paper was motivated by, on the one hand, a sense that automated chat sessions remain flat, without a dialogical dynamic that marks teaching and other human interaction in its ideal communicative moments. Aspects of this have been documented in prior work conducted by colleagues and I several years ago (Magee, 2026). In that work, we claimed to detect signs of dialogical modulation, achieved by a second 'voice' - another agent that, following a Freudian model, acted as the simulated interior superego to the main 'ego' agent interacting with a human (or another automated)

user. And, on the other, it was motivated by what might be seen as the essential *relational* condition of the teaching situation. Without a sense that the teacher is paying attention to a student, the 'learning' is purely performative. What is claimed here is that the learner must feel their learning is *recognized* by the teacher in a way that – while of course remaining gestural and simulated – must seem responsive.

This paper describes then the conditions under which an AI tutor could be constructed following cues from social and psychoanalytic theory. It does so in an unusual way: it involves working with Claude Code (Opus 4.5 and 4.6 models) to build the AI tutor, and then evaluate and document its performance. The data analyzed in *this* paper is then the final result of that process, another paper entirely authored (with the exception of one self-reflexive remark) by Claude Code itself.

This strange elliptical pattern involves along the way several meta-scholarly claims. Following Andrej Karpathy, it is tempting to describe this way of working glibly as a form of 'vibe scholarship'. I start off with a hunch, I prompt, I build an evaluation system, I simulate some exchanges, I review and prompt some more. This would also fall under traditional name like 'exploratory data analysis' and 'grounded theory' - approaches that foregrounded data-driven inquiry. But this new era of AI-assisted research is also not the same. It is less a question of inductive leaning than of a certain methodological abandonment to the vicissitudes of the language model.

This in turn begs the question of, in this particular relation of human to machine, who is leading who. The research follows the path of least resistance of the machine. But it is nothing - as the generated paper makes clear - without the human researcher's bias, and virtual insistence upon a result, being manifest. At times, the pattern of interaction follows that of a supervisor instructing a student only too eager to please. At others, it is the human - me - being led through the intricacies of an ablative study that always threatens, one prompt at a time, to spin off into exponential territory. And being surprised at the coy and semi-tragic admission of the AI - that recognitive learning is an ideal that it, the AI, can approach via simulation but never realize. This dialectical engagement enacts, methodologically, the thetic content of both papers.

And then what is to be made of the result - that modelling tutor-learner agent interactions on Freudian psychodynamics and, in particular, Hegelian recogntion actually results in consistently measured statistically? Aside from the trumping limitation that it is entirely synthetic, it is also seems plausible. A perverse revenge of the humanities (if Freud qualifies): critical theory as a master-prompting in an era of automated knowledge. At the very least such results could be read as an insistence upon traditions of inquiry otherwise superceded by the shimmering allure of technical novelty. A lesson from an historical teacher to students of the distracted present.

The paper proceeds from a short theoretical discussion of the teaching relationship to a rationale for how this particular evaluation simulates its digital equivalent. It then discusses the generated 'paper' that it shadows or mirrors. It concludes with implications of both papers - the human and the automated one - for a world in which its seems necessary to recognize the structural transformations happening to practices of teaching and research.

The Drama of Teaching

As Derrida writes in *The Postcard* (1980), the famous relationship between Socrates and Plato depicted in Matthew Paris' *Plato and Socrates*, was always open to interpretation. The classical interpretation was that Socrates the teacher spoke, while Plato the student faithfully transcribed the teaching. Paris' medieval image inverts this order: it is Socrates who writes, and as Derrida suggests, Plato who perhaps observes, directs or even produces the 'text' of Socrates. This scene of an originary moment of Western philosophy is also one of dialogical instruction: to think is also to involve an encounter between the one who knows and the one who needs to know. Though pacific enough, this image alludes to the inevitable drama involved in this encounter. Plato gesticulates; he has a point to prove. Socrates could be taking note, or could be ignoring his student, impatient to commit his own thoughts to the page. As Long (2014) suggests in a gloss on Derrida's discussion of the portrait, positioned behind Socrates Plato also appears in a position more commonly held by either angel or devil in medieval painting, including in others drawn by Paris. In this interpretation, as angel or devil, it is Plato who instructs, and Socrates who writes down the lesson. Derrida was keen to note even other, more whimsical possibilities: Plato, anachronistically but not out of character,

jumping on a tram or running off with a skateboard. These animated vignettes alude to all kinds of possible dramas unleashed by Paris' medieval cartoon.

In the context of the rise of AI tutors, this paper takes up this sense of the theatricality involved in teaching, and specifically, in the dynamic between teacher and student. It does so supported by two philosophers, discussed often by Derrida, who recognized the essentially and existentially dramatic nature of dyadic human relations: Hegel and Freud.

For Hegel, intersubjective relations are primordially established through the agonistic meeting between two subjects or minds. This meeting is critical to how consciousness becomes self-conscious. Paradoxically, it is only through encountering another self-consciousness that I myself can become properly conscious of myself as a self or subject. Until that point I am merely conscious, an evolving sensing, perceiving and understanding being who as yet is unable to move into a full life marked by desire and, eventually, recognition. It is this other who I perceive is both totally alien and, at the same time, something very like me: something also capable of sensing, perceiving and understanding. Yet for Hegel this meeting is not fortuitous, and leads to his description of the famous master-slave dialectic. The fraught encounter between these two consciousnesses, seeing something of themselves in the other and vice versa, is not some happy accident. It is necessary to greater learning, but it also has its risks. One party can die; this is an existential struggle. Or just as precipitously, from the point of view of the developing self-consciousness, the other party can gain mastery. To succumb to the power of that other is to be marked by servitude, incapable of realizing autonomy. Yet for Hegel, ironically, it is this mastery that marks the sudden arrestation of progress towards self-consciousness. Fortelling an endlessly repeated tragedy that, in more recent times, gets projected into human-machine affairs, it is the master who is discarded by history [^For some commentators, our recollection of this high drama owes more to Kojeve's extensive re-telling in].

Something of this same dramatic interlude appears in miniature in the novel teaching encounter. Introduced at the start of year or semester, the teacher presents themselves as a figure of authority. If they have not gained ascendency over the student, that is because their mastery has been acquired elsewhere, in the frequent 'defences' familiar to schools and the academy. As master of a subject they command attention from the student who, in a position of subservience sits – perhaps attentive, bored, or even rebellious, but even then, still recognizing the asymmetry of their relation to the one who teaches. Recognition is impossible without this prior acknowledgement of mastery. This of course does not describe every teaching reality, even prior to modern efforts to re-write this asymmetry according to an egalitatarian dynamic composed of facilitation and flipped classrooms. There are moreover all sorts of strategies for offsetting asymmetry: the teacher's humorous self-deprecation or selfless sublimation into the wonders of their field; the learner's heartfelt appreciation for the lessons, shown in their own progress in the discipline; or the joint struggle against the felt demands of a modern educational bureaucracy.

Still, something of this drama always remains, and at its heart lies the Hegelian concept of *recognition*. In Hegel's original presentation (2018), recognition was desired by both parties, an essential gesture toward self-consciousness. I cannot know myself unless I can see someone else knows me for what I am; without this corroboration or triangulation, I am plagued by doubt as to the the sort of thing I am. But an assymetrical relation mocks this desire. For the eventual servant is not recognized by the master, who after all cannot recognize what is beneath them. But conversely the master, who also craves recognition, has only the servant to recognize them. Like a comedic duo condemned to play out an eternal dialogue without resolution[^Beckett's plays feature many such dyads, Ponzo and Lucky from *Waiting for Godot* perhaps most famously], both roles circle about eachother in frustration. The opportunity for breakthrough comes not in the form of any resolution to this tension, but rather through the servant's immersion in the world of objects, which they in turn come to know and master. The lessons are autodidactic, unless it can be claimed that it is the master who manufactures, surreptitiously, in a gesture of subversive benevolence, the conditions for the servant's own learning.

What can be seen as the breaking through of this cycle in a teaching context is established by the collapse of this dynamic altogether. This is constituted when student and teacher recognize eachother via something like a temporal shift. When the student learns, the teacher seems themselves as they too once were, at the moment of apprehension. Conversely, in their struggle to comprehend, the student anticipates an eventual

Figure 1: Image

moment of mastery, when they too become like the teacher. The asymmetry of the present dissolves in this twin experience of recall and projection. The asymmetry is experienced, in other words, on both sides as *contingent*, a question of temporal ordering: were I you and you me, we might share this exact same moment in reverse, me now as teacher, you as student. This contingency itself falls away to reveal a more fundamental equivalance, and in this, we establish a mutual recognition.

For Freud, or perhaps more correctly, a Freudian reading of Hegel (see Kojeve), even this recognition remains fraught with antagonism. No encounter is ever purely dyadic. In the background lurk the ghosts of past encounters, other authority figures, other masters. In *The Ego and the Id*, Freud (Freud, 1961)employs the metaphor of a rider to represent the *egoic* mastery exercised over animalistic drives of the more powerful *id*. More ominously, the third interior character is the super-ego, an idealized figure resulting from early childhood parental identification. In later life, the super-ego is a 'precipitate' left over from this early project ideal, who continues to haunt the ego with precepts ('You ought to be such and such (like your father)' (Freud, 1961, p. 44)) and prohibitions ('You must not be such and such (like your father); that is, you may not do all that he does; many things are his prerogative' (Freud, 1961, p. 44)). Incidentally, Freud explicitly and immediately follows up this introduction of the super-ego by way of the Oedipal scene with acknowledgement of the reinforcement that comes by way of later repressive forces ('discipline, religious teaching, schooling and reading' (Freud, 1961, p. 45)). But what is significant here is the effective *doubling up* of both tutor and student relations. Leaving aside the presence of the Id or unconscious, it is as though the teaching encounter is also marked by secondary conversations. The teacher teaches the student, but the student's ego also experiences a separate *voice* of its super-ego, who may reinforce the tutor's lesson or, alternately, resist it.

An example of the latter: a teacher teaching the need to combat racism in the name of tolerance and equality might encounter resistance in the form of a student's scorn, barely masking much earlier lessons from a racist parent (typically but not always, as even Freud noted, a father). This resistance needs to be overcome by teachers employing transference techniques that echo (and also pre-date, in the case of 19th century novels) those of psychoanalysis.

This is no less true of the teacher, who through a Freudian lens also experiences an interior dynamic between a rational ego and ghosts of past ego-ideals, congealed into the form of conscience (Freud, 1961). Indeed the teacher must forever seek not only to exhibit authority in front of the student, but justify themselves before their own residues of other egoic figures. As Freud (1961, p. 52) puts it, 'Thus in the id, which is capable of being inherited, are stored up vestiges of the existences led by countless former egos; and, when the ego forms its super-ego out of the id, it may perhaps only be reviving images of egos that have passed away and be securing them a resurrection.' Translated into pedagogical practice, the teacher can be said to submit their teaching to the scrutiny of both student and, more terrifyingly, their own teachers who live on as internal supervisory echoes. Moreover, since the super-ego stems itself from early childhood, it eternally orients itself towards an infantalised form of the ego itself. Like an imposter syndrome that can never be dispelled, it refuses to believe the ego can ever amount to more than a child that always requires instruction. Somehow it mocks the teacherly ego, which is all the more defensive in its insistence upon its mastery. Perversely, it is only through the learner's recognition that the teacher can be reassured. Through that recognition the teacher's ego is able to insist upon the present-day reality – 'I am teaching and the student is learning' – against super-egoic scepticism. In this sense both learner and teacher are able to escape their respective ego-superego 'loops' only by transferring attention to the present reality of their intersubjective encounter - suspending these interior voices long enough to eventually re-program them.

'Recognitive-theoretic learning'

In theoretical terms this splicing together of exterior social relations and interior psychodynamics is not novel. Recent attempts to develop so-called 'recognitive-theoretic' (Stahl, 2013) ideas of a wider 'critical theory of society' (Honneth, 2014) have mined Hegel's conceptualization of recognition, and in the case of Honneth's work especially, have also sought to connect this social theory variant with psychoanalysis. Honneth's analysis is not uncritical of either Hegel or Freud, and has even looked to replace Freud with later object relations theory. To conclude this paper, it is helpful to revisit this analysis, since Honneth's wider aim also has applications in minimal social arrangement that consistutes the tutor-learner encounter.

For Honneth, critical theory has always needed an psychologizing account that can explain the paradigmatic twentieth century problem faced by socialist thinkers: why, when faced by the putatively obvious case for an equitable economic system of redistribution, do those who stand to benefit – the workers, the poor, and so on – consistently vote or act against their own interest? This is especially difficult to comprehend once extrinsic factors – fear of the struggle to death, ideological blinkers, and so on – fall away in modern democracies. To account for this requires a psychological theory capable of describing, as Honneth (and of course Freud (Freud, 1952) before him) puts it, the concept of negativity or negation. No other mainstream theory explains why social agents would consistently act against their apparent interest. In Freud's analysis, negation is a function that allows something repressed to be represented to consciousness precisel in the form of its denial. Projected into the social sphere, negation is further linked to the destructive and masochistic libidinal impulses that enable, for example, a member of the working class to simultaneously identify with and prostrate themselves before a master. In Adorno' social reworking of Freudian negation (Adorno, 1951), this curious dynamic explains the absence of progression to recognition. Rather than developing through its essential stages, Hegel's master/slave dialectic gets stuck in a kind of perpetual machine of domination and subservience that pleasures both parties. This pleasure is purely libidinal and coupled with aggressive tendencies; as Adorno notes, in its fascist incarnation in Nazi Germany, 'reference to love is almost completely excluded' and where it is mentioned, only with the 'epithet of "fanatical" through which even this love obtained a ring of hostility and aggressiveness against those not encompassed by it' (Adorno, 1951). Finally, even the ties binding leader to people is necessarily assymetrical: 'the leader can be loved only if he himself does not love' (Adorno, 1951).

Honneth's own reasons for seeking to move past this Freudian-Adorno analysis of social relations relates, it seems, precisely to a need to negate the negation, and find an alternate path to an socially emancipatory future founded upon mutual recognition across different social layers of the family, the community and the state. How does this recognition arise? Only through less destrutive possibilities, which for Honneth arrive through the elaboration and critique of the Freudian story through object relations theory. Here the possibility is instead that the social subject is able to grow to transfer attachment from inanimate objects to other people. This begins a period of hopefully permanent healthy egoic development, as attachments lose their infantile narcissistic and pleasure-seeking function and become integrated into a self-confident subject capable of maintaining a set of mutually constitutive social relations.

For this paper's case Honneth's argument is a muted one, as the simulated application of both recognition and Freudian architecture take place metaphorically and abstractly, via the equivalent of stage notes or prompts passed off-stage to the main characters (tutor/learner, ego/super-ego etc). High fidelity to the nuances of a particular interpretation is arguably less important that building a simulation structure that allows for more complex interactions – alongside the simulation of learning. The purpose of this lengthy detour hopefully becomes clear here. One hypothesis for the experimental success of both Hegelian and Freudian models is they re-situate the default dynamic of simulated tutor and learner from the asymmetry implicitly assumed even in contemporary 'flipped classroom' discourse – which still, at least in modern universities, depend upon quantified measures of student success – to one that instead stresses recogntion itself as part of a desirable pedaogical interaction.

Towards 'Machinagogy': Recognition and Acting Out

This brief characterization can too readily be seen as enacting its own deliberate dramatization, as though other more benign models might not be more credible, as well as being better supported by a rich empirical literature. This paper's aim though is not primarily one of defending a specific theoretical orientation. Instead it is to make more explicit the conceptual heritage that simulated pedagogy inherits, and not only from explicit models of education. For all its claims to novelty the field of machine learning continues to fall back upon comparable myths and idealizations of humanistic cognition and socialization. Reinforcement learning – the paradigm through which LLMs are able to 'align' with human values – derives from a behavioural and mechanistic model long since abandoned in education. Without naming Hegel or Freud, GANs – Generative Adversarial Models – re-stage the conflictual relationships described by both. As Minsky has noted in a provocatively title essay, Freud's psychic architecture was to prove influential in early AI experiments - especially those informed by cybernetics - and continues to echo, less explicitly, in the repressive effects

of reinforcement learning applied to LLMs. As generative AI has given rise to inevitably anthromorphic 'traits', researchers at Anthropic have released a series of papers that read as much like Freudian case studies of AI neuroses as they do technical papers. Replace 'large language model' with a human surrogate like 'infant' or 'patient', and the following Anthropic research paper titles could be taken from a 1920s psychoanalysis conference: 'Signs of introspection in large language models', 'Tracing the thoughts of a large language model', 'Alignment faking in large language models', 'From shortcuts to sabotage: natural emergent misalignment from reward hacking', 'Reasoning models don't always say what they think', 'Auditing language models for hidden objectives', 'Sycophancy to subterfuge: Investigating reward tampering in language models' (Anthropic, 2025a, 2025b; Chen et al., 2025; Denison et al., 2024; Greenblatt et al., 2024; MacDiarmid et al., 2025; Marks et al., 2025). Alongside flat anthropomorphic metaphor, machine learning has already long leaned into the structured intersubjective and interior registers of the humanistic disciplines.

This paper proposes an architecture for an AI tutor that borrows explicitly from Hegelian and Freudian analyses to condition the tutor's behavior. The purpose is to see - albeit within the constraints of purely synthetic environment - what a much-reduced and simplified instrumentation of these models can yield experimentally, in terms of greater quality of interaction, greater dialogical modulation and, ultimately, learning. In the first instance an agent representing the Tutor (and in certain configurations, the Tutor's *ego*) interacts with a learner (either human or, in the evaluations described here, another agent). That interaction is detailed in the tutor's prompt in one of two ways: either as a conventional interaction expected of any generic tutor, or as an interaction that strives towards Hegelian *recognition*.

Second, both tutor and learner can be modeled as single agent and multiagent systems. In the single agent, both learner and tutor are LLM agents interacting with eachother accordingly to a generic role-based prompt. In the second instance, following (Magee, 2026), either or both tutor and learner roles are modelled as a pair of LLMs agents, one of which represents the *ego*, the other, the *super-ego*. In this case when the ego receives a message from the other role, it first formulates a response, then plays the response *to the super-ego*. The super-ego then either approves or criticizes the message, wherupon the *ego* agent updates its response before forwarding it on to the other role.

These two additions have a purely demonstrative function: to see what happens when LLMs are configured to play out philosophical or psychoanalytic roles. But, as with (Magee, 2026), there is a pragmatic purpose: to see if these changes also can improve the performance of the tutor (and the learner, as one measure of the tutor's performance). The hypothesis is that LLM agents quickly devolve into a fixed and monotonous pattern without incentives to adapt and evolve. One method of coercing that adaptation is, again following (Magee, 2026), to stage an interior dialogue, with a super-egoic critic seeking to second-guess or check the (typically) breezy and flattering patter of the egoic teacher. Another is bring the concept of recognition right to the fore, in the form of prompt specifications that demand that the ego agent (with or without super-egoic interuption) seek to recognize and be recognized by the other agent.

Despite famous efforts at conciliation, this amalgam of Hegel and Freud is not itself necessarily a happy one, and in a certain sense, the implicit conflict between the two designs can itself play out in how, for example, learner ego and tutor super-ego monitor and seek to condition the tutor ego role. Moreover, by contrasting both interventions (the Freudian ego/super-ego again doubled, for tutor and learner role) with a generic baseline allows an ablative 2x2x2 experiment to be conducted: with/without learner and tutor super-ego roles, and with/without the recognitive prompt suggestions. A further LLM agent, powered by a commercial state-of-the-art model, then applies a rubric to judge both tutor and learner performance against evaluative rubrics.

Finally, different combinations of the LLMs themselves were selected for the different roles. All LLMs were requested through the OpenRouter service, as this provided a convenient means of using many models and configurations. The baseline model was Nemotron 3, an open source model operating free-of-charge on the OpenRouter service. Other combinations included use of other leading open source models, as well as commercial models by Anthropic and OpenAI. Adding model configurations added more complexity, but allowed another question to be posed: could combinations of free or lower cost models be convinced into meeting performance standards of other more expensive models with simpler or less directed parameters?

The entire experimental design and analysis was undertaken by applying a set of Claude Code + Opus 4.5 /

4.6 (hereafter 'Clopus') agents. These agents - distinct from those used in the experiment itself - were tasked with developing an evaluation harness and rubric; authoring initial prompts, with and without reference to recognition, for both sets of ego/super-ego roles; analysing results of the evaluation; writing up a Latex 'paper' with this analysis; and reviewing, critiquing and refining both the experimental design and write-up. To obtain meaningful results this involved a number of iterations, as the experimental design evolved and the various bugs were addressed. The 'data set' therefore includes all code, synthetic data produced through the experiments, prompts, interim notes, experimental outputs, analysis artefacts such as figures and tables, and the contents of the final 'paper' itself. What follows includes a critical 'review' of these papers, reflecting in part upon how the Claude Code agents also interpreted prompts to build this evaluation platform. The final 'paper' is included as an appendix, with references being made to this Claude Code artefact in the paper proper.

This unusual procedure reflects the similarly unusual affordances introduced by 'Clopus'. Alongside the erstwhile evaluation of the conceptual architecture, this study also discusses how this new combination operates as pseudo-researcher. In this orchestration, the human and machine 'researchers' routinely switch roles of 'learner' and 'teacher', signalling a shift that complicates not only an ontological boundary but also, at a certain level, reproduces the very drama – or more correctly, the comedy – that the research seeks to stage. What for now is an obscure promissory note will, by the Conclusion, hopefully be made clear.

Anatomy of a 'paper'

The most immediate implication of the study is that as far as prompt strategies for LLMs go, even classic theories from humanistic disciplines can provide architectural and rhetorical inspiration, with demonstrable results. The baseline prompts were still lengthy and sophisticated. However both the recognitive and multiagent facilities increased performance, at least as judged by another LLM. There are several ways to interpret these resuls. One could be that the LLM judge is 'persuaded' by the more sophisticated language of both injections. Another might be that this language also directs even smaller models like Nemotron 3 into a more philosophically-informed latent space in the model. In the case of both Hegel and Freud, it is not simply that their own writings are included in the training data; the vast secondary literature on both figures is captured too. This seems something like a citation effect: ask a LLM to behave like a tutor, and it will parrot a generic tutoring voice; ask a LLM to augment its tutelage with a more sophisticated framing, and it appears to raise its tone. It does appear at any rate that century-old theorizations of pedagogical scenarios are enough to orient model behavior into productive directions. Paradoxically, the most successful 'prompt engineers' might be those whose works have long since been canonized, and are now copied and widely cited in the LLM's training data. In relation to both research questions, the answer is a qualified 'yes', pending opportunity to validate with human users.

A second implication lies in the success of 'Clopus' acting as a co-researcher. To avoid concerns around authorship, the AI-authored 'paper' has been deliberated 'authored' without cross-over with this one, but it seems inevitable that the ability of systems like 'Clopus' will make more fragile the distinction between human and AI-generated research. Following the introduction of the term 'vibe coding' in 2025 (Meske et al., 2025; **2025LLMs?**), it is tempting to coin an equivalent phrase like 'vibe scholarship' to describe the conditions of this style of AI-augmented research. This idea would need to be distinguished from plagiaristic or even utilities like grammar checking. Instead it involves working with AI primarily as a coding agent, translating in this instance ideas from the humanities into prompts, implementing an evaluation framework, analysing results and, occasionally, using those results to refine its own framework. Its own expression of the framing and implications of the research, as expressed in the 'paper', serves to make explicit what the experimental design actually seeks to evaluate - and serves as a basis for follow-up suggestions to refine, complicate or simplify the design.

A third consideration, related to the second, involves the repetition at the level of method of precisely the object of investigation. What can be said of the dialectic and psychodynamic interplay between human and AI researcher? Who is Plato, who Socrates in this scenario? There is no clear answer, but also the ontological differences mean that at least currently, 'machine learning' is only ever metaphorical. Learning takes place in the transient space of a given dialogical frame, and once the curtain falls – and despite some limited memorization and personalization that lasts across frames - the lessons need to be repeated. In this sense

the lessons are at best circuitous, finding their way perhaps into much later training sets and the directives issued by engineers of foundational models.

Conversely human researchers – as figurative and often also actual tutors and learners – face complex dilemmas in these 'machinagogical' encounters. What once might have involved a million dollar grant, prolonged hiring decisions, technical trial and error, and several years of preparation and administration can now be channeled into several months of a consumer-grade subscription. Just the like the multidimensional space that underpins LLMs, LLMs agentive work activity can splinter off, often in parallel, to website design, evaluative methods, analysis of experimental data, and conceptual justification. That involves not only a compression of space and time, but a different dynamic between human and machine. This dynamic is a recognitive one, though only if 'recognition' is thought, in a post-Hegelian gesture, as also permissible across hard ontological lines. I do not need to recognize this kind of machine as another self-consciousness, only as a mechnical device capable of simulating a consciousness, capable of grappling with a supple and iterating set of instructions.} }

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(**misc?**){libguidesumnedu2026Characteristicsgood, title = {Characteristics of a good research question - Literature Searching - Research Guides at University of Minnesota Minneapolis}, author = {libguides.umn.edu}, journal = {libguides.umn.edu}, url = {https://libguides.umn.edu/c.php?g=1337354&p=9854773}, abstract = {Overview and tips on how to conduct a literature search.} }

(**article?**){Prompter1925Thiswild, title = {This is wild Multi-agent LLMs are starting to think together.A new paper "Emergent Coordination in Multi-Agent Languag…}, author = {Alex Prompter}, year = {1925}, journal = {X (formerly Twitter)}, url = {https://x.com/alex_prompter/status/1978024501725319556}, abstract = {This is wild Multi-agent LLMs are starting to think together.A new paper"Emergent Coordination in Multi-Agent Language Models" just dropped, and it's mind-blowing.Researchers found that when you connect multiple LLMs with light feedback (and no direct communication),… pic.twitter.com/PUWgdx6NhJ— Alex Prompter ((**alex_prompter?**)) October 14, 2025} }

(**misc?**){sainingxie1925threeyears, title = {three years ago, DiT replaced the legacy unet with a transformer-based denoising backbone. we knew the bulky VAEs would …}, author = {Saining Xie ((**sainingxie?**))}, year = {1925}, date = {1925-11-08}, journal = {X (formerly Twitter)}, url = {https://x.com/sainingxie/status/1977936710135669130}, abstract = {three years ago, DiT replaced the legacy unet with a transformer-based denoising backbone. we knew the bulky VAEs would be the next to go – we just waited until we could do it right.today, we introduce Representation Autoencoders (RAE).» Retire VAEs. Use RAEs. (1/n) pic.twitter.com/NrQcW37hYL— Saining Xie ((**sainingxie?**)) October 14, 2025} }

(**article?**){Citation2026Preprint, title = {Preprint PDF}, url = {http://arxiv.org/pdf/2507.21928v1} }

(**article?**){Meske2025VibeCoding, title = {Vibe Coding as a Reconfiguration of Intent Mediation in Software Development: Definition, Implications, and Research Agenda}, author = {Christian Meske and Tobias Hermanns and Esther von der Weiden and Kai-Uwe Loser and Thorsten Berger}, year = {2025}, date = {2025-07-29}, doi = {10.48550/arXiv.2507.21928}, url = {http://arxiv.org/abs/2507.21928}, abstract = {Software development is undergoing a fundamental transformation as vibe coding becomes widespread, with large portions of contemporary codebases now being AI-generated. The disconnect between rapid adoption and limited conceptual understanding highlights the need for an inquiry into this emerging paradigm. Drawing on an intent perspective and historical analysis, we define vibe coding as a software development paradigm where humans and generative AI engage in collaborative flow to co-create software artifacts through natural language dialogue, shifting the mediation of developer intent from deterministic instruction to probabilistic inference. By intent mediation, we refer to the fundamental process through which developers translate their conceptual goals into representations that computational systems can execute. Our results

show that vibe coding reconfigures cognitive work by redistributing epistemic labor between humans and machines, shifting the expertise in the software development process away from traditional areas such as design or technical implementation toward collaborative orchestration. We identify key opportunities, including democratization, acceleration, and systemic leverage, alongside risks, such as black box codebases, responsibility gaps, and ecosystem bias. We conclude with a research agenda spanning human-, technology-, and organization-centered directions to guide future investigations of this paradigm.}, note = {arXiv:2507.21928 [cs]}, keywords = {Computer Science - Artificial Intelligence, Computer Science - Human-Computer Interaction, Computer Science - Software Engineering} }

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(**article?**){Radford2018Improvinglanguage, title = {Improving language understanding by generative pre-training}, author = {Alec Radford and Karthik Narasimhan and Tim Salimans and Ilya Sutskever}, year = {2018}, date = {2018}, url = {https://www.mikecaptain.com/resources/pdf/GPT-1.pdf}, note = {Publisher: San Francisco, CA, USA} }

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(**article?**){Kaloka2024ImprovementExecutive, title = {Improvement of Executive Function Through Cognitively Challenging Physical Activity with Nonlinear Pedagogy In Elementary Schools}, author = {Pasca Tri Kaloka and Soni Nopembri and Yudanto Yudanto and Gunathevan Elumalai}, year = {2024}, date = {2024-01-01}, journal = {Retos}, volume = {51}, pages = {673-682}, doi = {10.47197/retos.v51.101024}, url = {https://revistaretos.org/index.php/retos/article/view/101024}, abstract = {This study investigated the effects of cognitively challenging with nonlinear pedagogical games on the executive functions and situational interests of students in physical education. Participants were 145 fifth and sixth-grade students (74 boys, 71 girls) from four elementary schools. This acute investigation, which included a physical education session, utilized a quasi-experimental design with four groups, repeated measures, and cross-over. After the post-test, one fifth- and one sixth-grade class from each school was randomly assigned to Group 1 with cognitively challenging nonlinear pedagogy games, Group 2 with a session for teaching soccer skills, Group 3 with a session for teaching track and field skills, and Group 4 control group, all the group with cognitively challenging physical with activity nonlinear pedagogy games. Included were pre- and post-test measures of executive functioning and a post-test assessment of situational interest. Students in Group 1 who engaged in cognitively challenging physical activity with nonlinear pedagogy games increased their executive function scores ($t(39) = -19.75$, $p < 0.001$, $d=1.82$) more than students who participated in sessions with group 2 soccer ($t(33) = -3.14$, $p=0.005$), group 3 track and field skills ($t(38) = -0.98$, $p=0.325$) and those group 4 the control group ($t(32) = -0.68$, $p=0.523$). The positive effects on the executive functions of students were duplicated when the cognitively demanding physical activity with nonlinear pedagogy games session was administered to the control group on the waiting list Students who participated in the soccer skills session showed some improvements in their executive functions. Students who participated in cognitively challenging physical nonlinear games scored higher on novelty than those who participated in soccer or track and field. These findings confirm the efficacy of cognitively challenging physical activity using nonlinear games in stimulating the executive functions of students in physical education. Key words: physical activity, nonlinear pedagogy, situational interest, cognitive engagement, design fluency}, keywords = {Actividad

física, cognitive engagement, compromiso cognitivo, design fluency, fluidez en el diseño., interés situacional, nonlinear pedagogy, pedagogía no lineal, physical activity, situational interest} }

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(**article?**){Posner2007ResearchAttention, title = {Research on Attention Networks as a Model for the Integration of Psychological Science}, author = {Michael I. Posner and Mary K. Rothbart}, year = {2007}, date = {2007-01-01}, journal = {Annual Review of Psychology}, volume = {58}, number = {1}, pages = {1-23}, doi = {10.1146/annurev.psych.58.110405.085516}, url = {https://www.annualreviews.org/doi/10.1146/annurev.psych.58.110405.085516}, abstract = {As Titchener pointed out more than one hundred years ago, attention is at the center of the psychological enterprise. Attention research investigates how voluntary control and subjective experience arise from and regulate our behavior. In recent years, attention has been one of the fastest growing of all fields within cognitive psychology and cognitive neuroscience. This review examines attention as characterized by linking common neural networks with individual differences in their efficient utilization. The development of attentional networks is partly specified by genes, but is also open to specific experiences through the actions of caregivers and the culture. We believe that the connection between neural networks, genes, and socialization provides a common approach to all aspects of human cognition and emotion. Pursuit of this approach can provide a basis for psychology that unifies social, cultural, differential, experimental, and physiological areas, and allows normal development to serve as a baseline for understanding various forms of pathology. D.O. Hebb proposed this approach 50 years ago in his volume Organization of Behavior and continued with introductory textbooks that dealt with all of the topics of psychology in a common framework. Use of a common network approach to psychological science may allow a foundation for predicting and understanding human behavior in its varied forms.} }

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(**book?**){Stea2022PhrenologyNeuroscience, title = {Phrenology and Neuroscience}, author = {Jonathan N. Stea and Tyler R. Black and Stefano I. Di Domenico}, year = {2022}, date = {2022}, publisher = {Routledge}, abstract = {Co-authored by clinical psychologists and a psychiatrist, this chapter examines claims related to phrenology. It includes a section on pop culture and public perception. The chapter describes a recent investigation into phrenology that uses 21st century scientific methods to investigate this historic 19th century practice. The chapter makes the case that phrenology helped to set the stage for the future scientific understanding and investigations into the brain and neuroscience. The chapter concludes by describing the historical use of phrenology to promote racist ideas.}, note = {Num Pages: 11} }

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Z2wC&oi=fnd&pg=PA284&dq=Attentional+control+and+self-regulation+in+early+development&ots=5ltwH26qYm&sig=
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(**misc?**){Leavy2025FullTranscript, title = {Full Transcript: Google Engineer Talks}, author = {Elliot Leavy}, year = {2025}, date = {2025-03-20}, url = {https://www.aidataanalytics.network/data-science-ai/news-trends/full-transcript-google-engineer-talks-to-sentient-artificial-intelligence-2}, abstract = {A Google engineer has been placed on leave for violating company confidentiality after he claimed that Google's Language Model for Dialogue Applications (LaMDA)} }

(**article?**){Marchetti2014Brainintersubjectivity, title = {Brain and intersubjectivity: a Hegelian hypothesis on the self-other neurodynamics}, author = {Igor Marchetti and Ernst H. W. Koster}, year = {2014}, date = {2014-01-24}, journal = {Frontiers in Human Neuroscience}, volume = {8}, doi = {10.3389/fnhum.2014.00011}, url = {https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2014.00011/full}
abstract = {IntroductionHuman beings live in a social context, where interacting with others is crucial for survival, and having a clear representation of both the "self" and the "other" is needed (Baumeister, 2011). Interestingly, neuroscience has traditionally adopted a Cartesian perspective by which the self is a solipsistic and self-sufficient unit, fundamentally unrelated to the other individuals' representation. Since the last decade, however, neuroscience is increasingly promoting research about how the "self" and the "other" are processed and represented at both intra- (Decety and Sommerville, 2003; Denny, Kober, Wager, and Ochsner, 2012; Molnar-Szakacs and Uddin, 2013; Wagner, Haxby, and Heatherton, 2012) and inter-brain level (Dumas, Martinerie, Soussignan, and Nadel, 2012; Dumas, Nadel, Soussignan, Martinerie, and Garnero, 2010). Nevertheless, the specific way whereby neural self-other representations co-occur and exert influence on each other in order to promote higher-order functions necessary for social functioning remains largely unclear. We propose that the philosophical theory of intersubjectivity (Hegel, 1807/1977) could integrate neuroscience findings and, in turn, shed new light on the self-other dynamics at neurocognitive level. Hegel and self-other dynamicsPhilosophy has intensively studied the interaction between the self and other, terming sometimes their interplay as intersubjectivity. In 1807, the German philosopher, Georg Wilhelm Friedrich Hegel, published the Phenomenology of Spirit (Hegel, 1807/1977), where he thoroughly investigated the progressive steps that from a relatively simple form of thinking, such as consciousness, lead to higher-order functions, such as self-consciousness. According to Hegel, consciousness is the mental function that accounts for ordered experience and which provides us with a unified experience of reality, instead of a chaotic flow of meaningless information. By means of consciousness, we are therefore able to represent and process in a meaningful way the stimuli that fall in our phenomenal space. Recent theoretical efforts have associated this crucial mental function with specific brain dynamics: rest-stimulus interaction at the level of cortical and subcortical midline regions (Northoff, 2012). Compared with earlier philosophers, Hegel took the notion of consciousness one step further. In line with his idealistic assumptions, he proposes that the mind actively "constructs" the objects of our knowledge, with no need for the latter to rely on cognitively inaccessible external entities. Earlier thinkers, such as Immanuel Kant, had indeed argued that these founding external entities were not comprehensible to our mind (i.e. thing-in-itself or noumenon) (Kain, 2005). Although his theory probably qualifies Hegel as one of the first radical constructivists, it also raises a fundamental question. If neither based on external reality nor on itself, on what is our self-consciousness based? Hegel (1807/1977) clearly answers that "Self-consciousness exists in and for itself when, and by the fact that, it so exists for another; that is, it exists only in being acknowledged" (paragraph 178) (Hegel, 1807/1977). Three elements are noteworthy here. First, Hegel proposes a condition of necessity in the self-other dyad, with each pole needing to be recognized by the other one (i.e. mutual recognition or intersubjectivity). Second, both the self and the other maintain their specific identity without merging into an undifferentiated matrix (Coelho and Figueiredo, 2003), in that the other is recognized as a separate subjectivity with whom a shared subjective state is possible (i.e. recognition of the other). Third, the self, at least in part, exists insofar as being constructed by the other that recognizes it back (i.e. recognition of the self). Without the other, the self lacks a fundamental reference to contrast itself with, and, in turn, it is bound not to emerge in the biological and psychological context (Baumeister, 2011). Importantly, a simple object fundamentally lacks self-consciousness and as such it is not eligible to recognize one's self as this would lead the latter to become "a motionless tautology of I am I" (paragraph 167) (Hegel, 1807/1977). In other words, if the other's recognition is lacking, I can experience events and objects (i.e. consciousness), but I cannot experience myself as a self-conscious agent (i.e. self-consciousness). From the intersubjective

perspective, the "other" is necessary. Not only must the other be physically present with its own body (Mead, 1934/2009; Russon, 1997), but the other must also recognize the subject as an intentional and self-conscious self (Kain, 2005). In order to process such fundamental but complex input, the brain is expected to be properly equipped to detect recognition by others. Intersubjectivity, Default Mode Network, and Mirror Neuron SystemHow may Hegelian intersubjectivity inform our knowledge of the brain and, in turn, account for our social functioning? Intersubjectivity could shed light on recent neuroimaging findings by integrating two of the most active research lines in neuroscience, the Mirror Neurons System (MNS) and the Default Mode Network (DMN).In the social context, the perception of a meaningful and goal-directed action leads to specific neural activation, regardless whether the action is performed by the subject itself or it is observed being performed by another actor (Iacoboni et al., 1999). The MNS has indeed been consistently associated with this function whereby the other's instrumental behavior is neuronally represented in the observer by means of a motor resonance mechanism. In other words, a significant gesture is encoded also in the observer's brain within its own motor schemata and this helps qualify the person of the observer as a (potentially) active and goal-directed actor. On the contrary, immobile objects or aimless actions do not elicit a similar response in the MNS (Preston and De Waal, 2002). Interestingly, already in the 1930's George Herbert Mead, a sociologist deeply influenced by Hegel's theory, stressed the importance of this "conversation of gestures" between individuals as the first step leading to higher levels of self-consciousness (Markova, 1990; Mead, 1934/2009). The MNS can therefore be considered as a neural tool that bridges the gap between the self and the other at the level of lower-order physical representation (Rizzolatti and Sinigaglia, 2010). The DMN, a neural network highly active during rest, has been associated with psychological representing of others, that is mentalizing (Schilbach et al., 2012; Van Overwalle and Baetens, 2009). This cognitive ability to adopt someone else's perspective and internally represent mental states of others (i.e. beliefs, intentions, and goals) implies that we subsume the cognitive representation of other's mental activity into ours to some extent. Interestingly, while the MNS relies on the perception of other's embodied actions, mentalizing requires the ability to extract and understand goals and intentions of others. The latter implies that the other is recognized as an agent. Therefore, the DMN seems to subserve the mechanism through which the other psychologically resonates into one's mind as an intentional being (Molnar-Szakacs and Uddin, 2013). As intersubjectivity is emerging as a promising perspective in neuroscience (Allen and Williams, 2011), we contend that it can meaningfully integrate these findings. Our Hegelian model suggests that it is the long-term interaction between the pre-reflective "I" (i.e. consciousness) and the other's representation, both at physical (Mead, 1934/2009; Russon, 1997) and psychological (Kain, 2005) level, that gives rise to a self-reflective "I" (i.e. self-consciousness). Hence, a self-conscious individual is capable not only of processing internal/external stimuli, but also of actively reflecting upon herself, which creates a network of information related to the self, such as attributes, beliefs, and traits (Baumeister, 2011).In neurocognitive terms, we could say that self-awareness is the product of interactions between both lower- and higher-order functions, such as motor resonance and mentalizing subserved by the MNS and DMN, respectively. Crucially, this neural cooperation has been recently confirmed by a meta-analysis showing that self-specific activity emerges as interaction between the DMN (i.e., perigenual anterior cingulate cortex [pACC] and posterior cingulate cortex [PCC]), and MNS (i.e., left anterior insula [lAI] and right inferior frontal gyrus [rIFG]) (Qin and Northoff, 2011). Moreover, mentalizing about the self and other is associated with similar neurocircuitry, yet, the brain seems to be capable of distinguishing between them without equating one to the other (Lombardo et al., 2010). In sum, Hegel's model of mutual recognition converges with neuroscience findings in that in order to understand ourselves we must rely on the same mechanisms that we use for understanding others. Self and other representations are strictly bond without one dissolving into the other (Coelho and Figueiredo, 2003; Lombardo et al., 2010). Notwithstanding this, the representation of others' minds and actions shapes our own mind, as both motor resonance mechanism (i.e., MNS) and mentalizing (i.e., DMN) are necessary to support and maintain the self-reflective self (Lombardo et al., 2010; Qin and Northoff, 2011, Sandrone, 2013). It is noteworthy that a similar hypothesis has been recently proposed by Timmermans, Schilbach, Pasquali, and Cleeremans (2012). The authors propose that a set of neurobiological prediction-based mechanisms support our constant attempt to model other's mind and the related social interaction. Importantly, the very same mechanism is proposed to be crucial for developing self-consciousness too. The master-slave dynamics in the brainRepresentation of the self and other is associated to a large extent with the same neural circuitry (Lombardo et al., 2010), and mutual recognition of self and other is required (Hegel, 1807/1977; Kain, 2005). However, this double bind

implies a fragile equilibrium between the two components of the dyad. What if, for instance, the self fails to recognize the other? What if mutual recognition is deficient (Williams, 1997)? Hegel (1807/1977) addresses this point by introducing the famous literary example of the master-slave dynamic, as follows. Given a couple of peers, one of the two could desire to undermine equilibrium by overpowering the other. Hence, one would become the "master" while the other becomes the "slave". However, the situation is bound to turn out paradoxical. By not considering the other a full self-consciousness (i.e. objectification), the master prevents the slave from recognizing it back. Consequently, the master hinders the process of mutual recognition upon which it itself relies in order to emerge as a self-conscious individual. In cognitive terms, we could say that the individual enhances the threshold to receive the recognizing input that could structure it as self-conscious. From this, several hypotheses derive. First, objectifying the other (i.e., negating its status of human being) leads, to a certain extent, also to a self-objectification, as the source of recognition is now lacking. Second, objectifying the other is expected to impact on the DMN and/or the MSN, the neurobiological systems maintaining self-consciousness. Preliminary findings support these hypotheses. First, being subjected to ostracisim (i.e., seeing the status of human being negated) leads the victims to judge both themselves and the ostracizer(s) less human (Bastian and Haslam, 2010). Second, actively engaging in ostracism makes the ostracizer feel less related to human beings (Legate, DeHaan, Weinstein, and Ryan, 2013). Third, a recent study showed that inducing participants to consider other human beings as objects leads to decreased activity of the DMN (Jack, Dawson, and Norr, 2013).Concluding remarksHegel's theory of intersubjectivity seems capable of shedding new light on the complex interaction between the self and the other at both neural and cognitive level. The interplay between the DMN and the MNS in supporting self-awareness may indeed be interpreted fruitfully by reintroducing the concept of mutual recognition. Importantly, new hypotheses can be derived from the classic work of Hegel in order to better account for the interaction between DMN and MNS in both healthy and clinical samples. For instance, major psychopathologies characterized by abnormal self-other dynamic, such as schizophrenia, autism, and depression (Billeke, Boardman, and Doraiswamy, 2013; Crespi and Badcock, 2008; Gallese, Rochat, and Berchio, 2013; Mehta, Basavaraju, Thirthalli, and Gangadhar, 2012; Northoff, 2007), represent key areas to test the explanatory power of the intersubjective theory. Finally, our neurophilosophical framework could be fruitfully integrated in a new branch of social neuroscience, namely the two-person neuroscience (Schilbach et al., 2013). By focusing on the neurocognitive basis of the interaction between two individuals, two-person neuroscience is developing new experimental paradigms and innovative methods to analyze real-time two-brain interplay, such as simultaneous neuroimaging recording or, so-called, hyperscanning (Babiloni and Astolfi, 2012; Dumas, 2011). In sum, we argue that the integration of our Hegelian neurophilosophical approach and two-person neuroscience holds promise to convey innovative future perspectives in the field of social neuroscience.}, note = {Publisher: Frontiers}, keywords = {Default Mode Network, Dehumanization, Hegel, intersubjectivity, mirror neuron system, self concept, self-consciousness} }

(**book?**){Taylor1983Hegelsphilosophy, title = {Hegel's philosophy of mind}, author = {Charles Taylor and Guttorm Fløistad}, year = {1983}, date = {1983}, pages = {133-155}, publisher = {Springer Netherlands}, url = {https://doi.org/10.1007/978-94-009-6932-2_6}, abstract = {As with any highly systematic body of thought, Hegel's philosophy of mind can be reconstructed from many perspectives. Each one gives us something though some are more illuminating than others. But I believe that it is particularly illuminating to see Hegel's Philosophy of mind though the perspective of his philosophy of action. This is of course hardly a surprising doctrine. Mind (Geist) for Hegel is thoroughgoing activity (Tätigkeit). But I think the insights from this perspective can be enlivened made more penetrating if we relate Hegel's thought to a set of perennial issues that have been central to the philosophy of action in modern times. This is what I want to attempt in this paper.}, note = {DOI: 10.1007/978-94-009-6932-2_6}, keywords = {Causal Theory, Mental Life, Qualitative Distinction, Qualitative Theory, Seventeenth Century} }

(**article?**){Hansen2024Fromattention, title = {From attention economy to cognitive lock-ins}, author = {Morten Hansen}, year = {2024}, date = {2024-09-01}, journal = {Big Data & Society}, volume = {11}, number = {3}, pages = {20539517241275878}, doi = {10.1177/20539517241275878}, url = {https://doi.org/10.1177/20539517241275878}, abstract = {The economic logic of the attention economy is frequently used to critique and respond to the dangers of unfettered technological expansion, including nascent platforms and products powered by generative artificial intelligence. This commentary warns that while large parts of the internet have been financed through such business models, there is no guarantee

that emerging generative artificial intelligence products will be commercialized in this way too. Instead, I argue, we must look beyond the attention economy to predict the future of monetization of an industry already mired in anti-competitive practices. Using popular large language models such as OpenAI's ChatGPT as a case, I discuss how some platforms are developing computational dependencies between technology and their users. I propose the term 'cognitive lock-in' to help us unpack the implications of such technological dependencies, and redirect the study of this nascent business model.}, note = {Publisher: SAGE Publications Ltd} }

(**article?**){Citation2026AvailableVersion, title = {Available Version (via Google Scholar)}, url = {https://www.culturemachine.net/wp-content/uploads/2019/01/465-973-1-PB.pdf} }

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}

(**article?**){Petersen2012AttentionSystem, title = {The Attention System of the Human Brain: 20 Years After}, author = {Steven E. Petersen and Michael I. Posner}, year = {2012}, date = {2012-07-21}, journal = {Annual Review of Neuroscience}, volume = {35}, number = {1}, pages = {73-89}, doi = {10.1146/annurev-neuro-062111-150525}, url = {https://www.annualreviews.org/doi/10.1146/annurev-neuro-062111-150525}, abstract = {Here, we update our 1990 Annual Review of Neuroscience article, "The Attention System of the Human Brain." The framework presented in the original article has helped to integrate behavioral, systems, cellular, and molecular approaches to common problems in attention research. Our framework has been both elaborated and expanded in subsequent years. Research on orienting and executive functions has supported the addition of new networks of brain regions. Developmental studies have shown important changes in control systems between infancy and childhood. In some cases, evidence has supported the role of specific genetic variations, often in conjunction with experience, that account for some of the individual differences in the efficiency of attentional networks. The findings have led to increased understanding of aspects of pathology and to some new interventions.} }

(**article?**){Vaswani2017Attentionneed, title = {Attention is all you need}, author = {Ashish Vaswani and Noam Shazeer and Niki Parmar and Jakob Uszkoreit and Llion Jones and Aidan N. Gomez and Łukasz Kaiser and Illia Polosukhin}, year = {2017}, date = {2017}, journal = {Advances in neural information processing systems}, volume = {30}, url = {https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html} }

(**book?**){Hegel1977PhenomenologyMiller, author = {Hegel, Georg Wilhelm Friedrich}, title = {Phenomenology of Spirit}, year = {1977}, publisher = {Clarendon Press}, address = {Oxford}, translator= {Miller, A. V.}, note = {With analysis of the text and foreword by J. N. Findlay.}, isbn = {0198245300} }

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